

ATMOSPHERE – EVOLUTION OF THE ATMOSPHERE

KEY VOCABULARY

Early atmosphere: mainly CO₂ from intense volcanic activity.

Photosynthesis: algae and plants made O₂ and removed CO₂.

Oceans: formed as water vapour condensed; dissolved CO₂.

Today: about 78% nitrogen and 21% oxygen.

COMMON MISCONCEPTIONS

~~—Oxygen was always plentiful.~~

✓ Oxygen built up only after photosynthesis evolved.

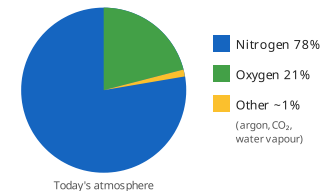
~~—The early air was mostly oxygen.~~

✓ It was mostly carbon dioxide.

~~—CO₂ levels never changed.~~

✓ CO₂ fell as it dissolved in oceans and made rocks.

ATMOSPHERE TODAY



1 EARLY GAS

Name the main gas in the **early atmosphere** and its source. [2 marks]

2 OXYGEN'S ORIGIN

Explain how oxygen was added to the atmosphere. [2 marks]

3 REMOVING CO₂

Describe two ways carbon dioxide was removed from the air. [2 marks]

4 TODAY'S AIR

State the approximate percentages of nitrogen and oxygen today. [2 marks]

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

Early ____ released CO₂; later ____ produced oxygen.

photosynthesis

volcanoes

oxygen

carbon dioxide

6 EVIDENCE

Suggest why theories about the early atmosphere are uncertain. [1 mark]

TIP: Early air: mostly CO₂. Today: ~78% N₂, ~21% O₂ — oxygen came from photosynthesis.

✓ Think: How did the formation of oceans help lock away carbon dioxide?